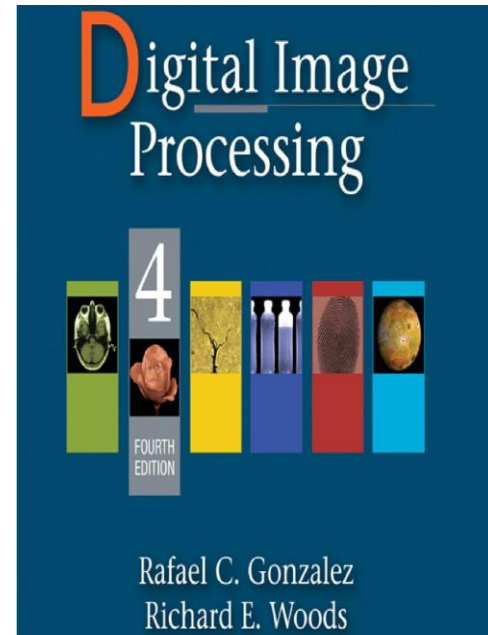
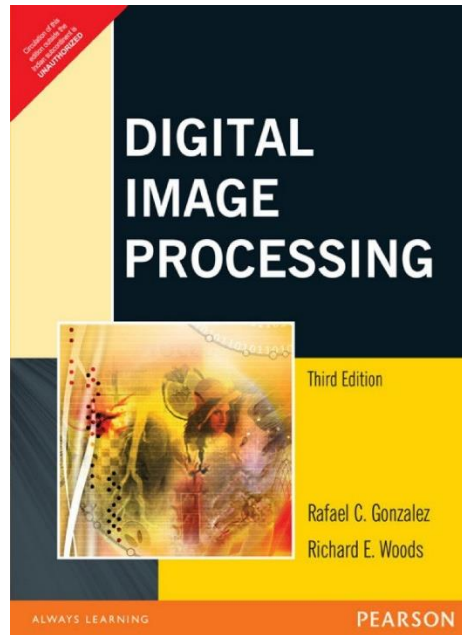
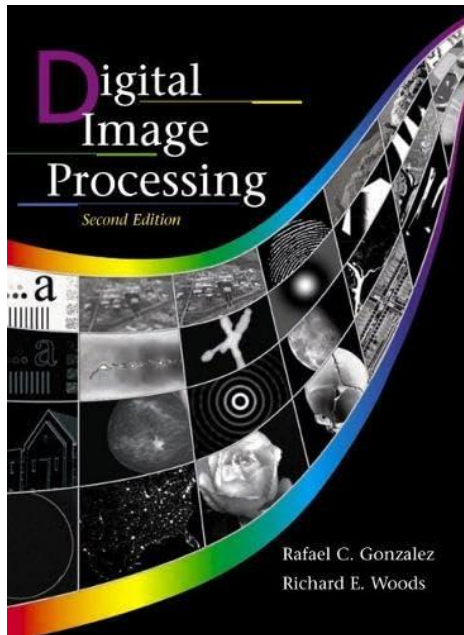


Digital Image Processing

Course Teacher: Prof. Dr. Rameswar Debnath

Ref. Book: Rafael. C. Gonzalez and Richard E. Woods,
Digital Image Processing, 2nd/3rd ed., Pearson
Education (Singapore) Pte. Ltd.

Recommended Book



1 Introduction

- Digital image processing focuses on two major tasks:
 - Improvement of pictorial information for human interpretation.
 - Processing of image data for storage, transmission, and representation for autonomous machine perception.

Introduction

- Images play the single most important role in human perception.
- One picture is worth more than ten thousand words.
- Computer vision is a field of artificial intelligence that trains computers to interpret and understand the visual world.
- Objective of computer vision is to emulate human intelligence.

Image Overview

- An image may be defined as a two-dimensional function, $f(x, y)$, where x and y are *spatial* (plane) coordinates, and the amplitude of f at any pair of coordinates (x, y) is called the *intensity* or *gray level* of the image at that point.
- When x , y , and the amplitude values of f are all finite, discrete quantities, we call the image a *digital image*.

Digital Image Processing

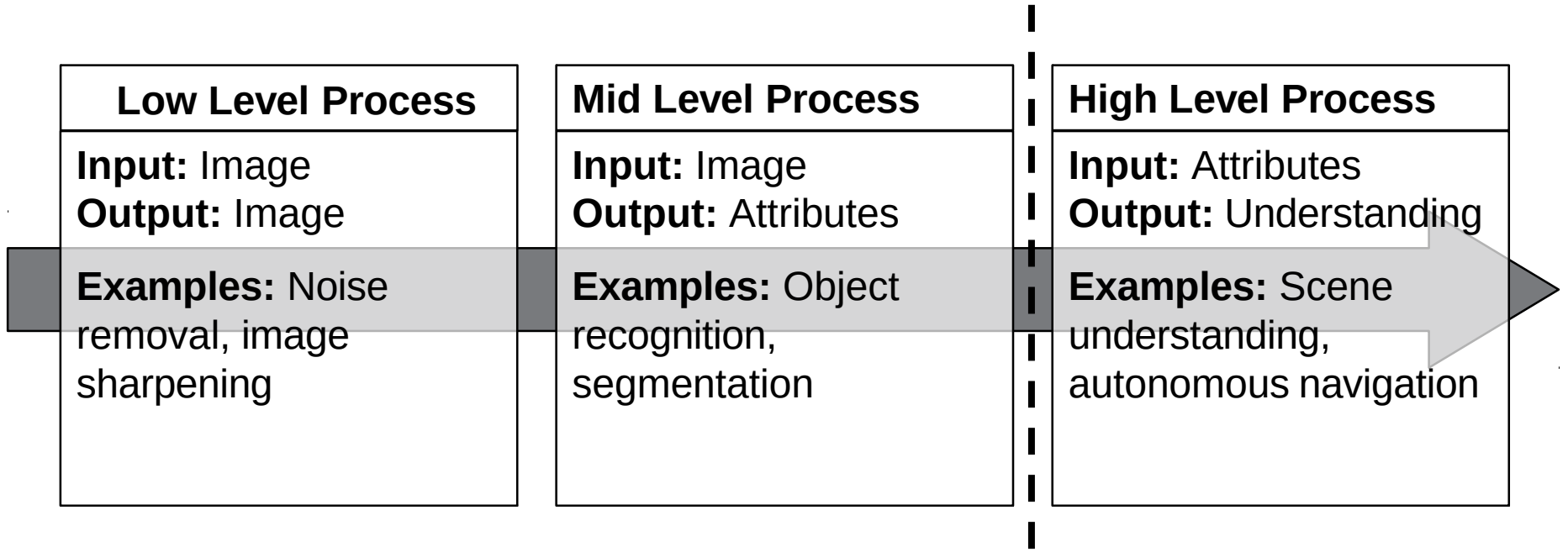
- Digital Image Processing (DIP) is a software which is used to manipulate the digital images by the use of computer system.
- It is also used to enhance the images, to get some important information from it.
- For example: Adobe Photoshop, MATLAB, etc.
- Digital image processing is one of the fastest growing industry which affects everyone's life.

Types of Image Processing

- Three types of computerized processes in digital image:
 - Low-level process
 - Mid-level process
 - High-level process
- Low-level processes involve primitive operations such as image preprocessing to reduce noise, contrast enhancement, and image sharpening.
- A low-level process is characterized by the fact that both its inputs and outputs are images.

Types of Image Processing

- Mid-level processing on images involves tasks such as segmentation (partitioning an image into regions or objects), description of those objects to reduce them to a form suitable for computer processing, and classification (recognition) of individual objects.
- A mid-level process is characterized by the fact that its inputs generally are images, but its outputs are attributes extracted from those images (e.g., edges, contours, and the identity of individual objects).
- Higher-level processing involves “making sense” of an ensemble of recognized objects and performing the cognitive functions normally associated with vision.



Applications of Digital Image Processing

- Image sharpening and restoration
- Medical field
 - Gamma ray imaging
 - PET scan
 - X Ray Imaging
 - Medical CT
 - UV imaging
 - Ultrasound imaging

Applications of Digital Image Processing

- Remote sensing
- Transmission and encoding
- Machine/Robot vision
- Color processing
- Pattern recognition
- Video processing
- Microscopic Imaging
- Others

Fundamental Steps in DIP

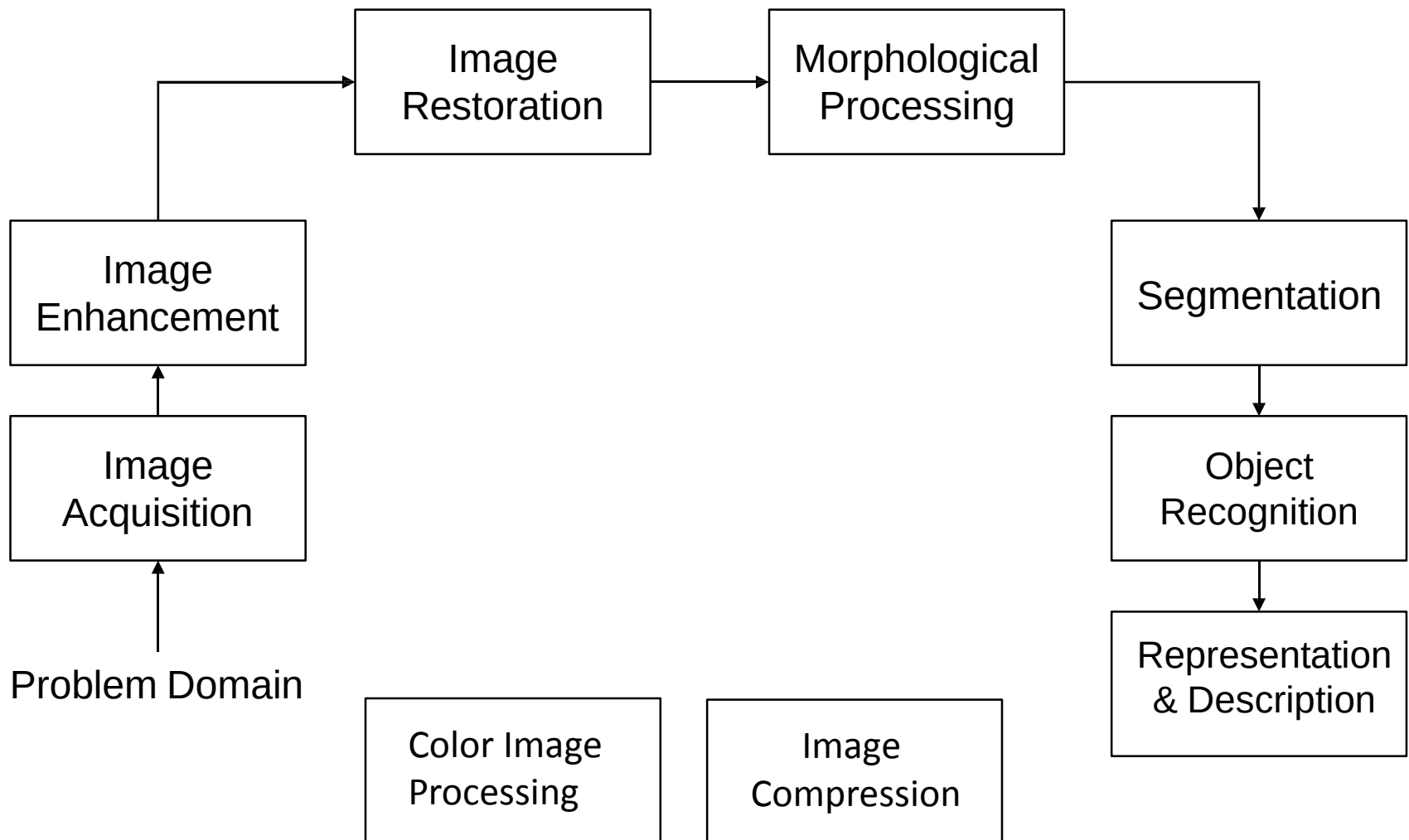


Image Acquisition

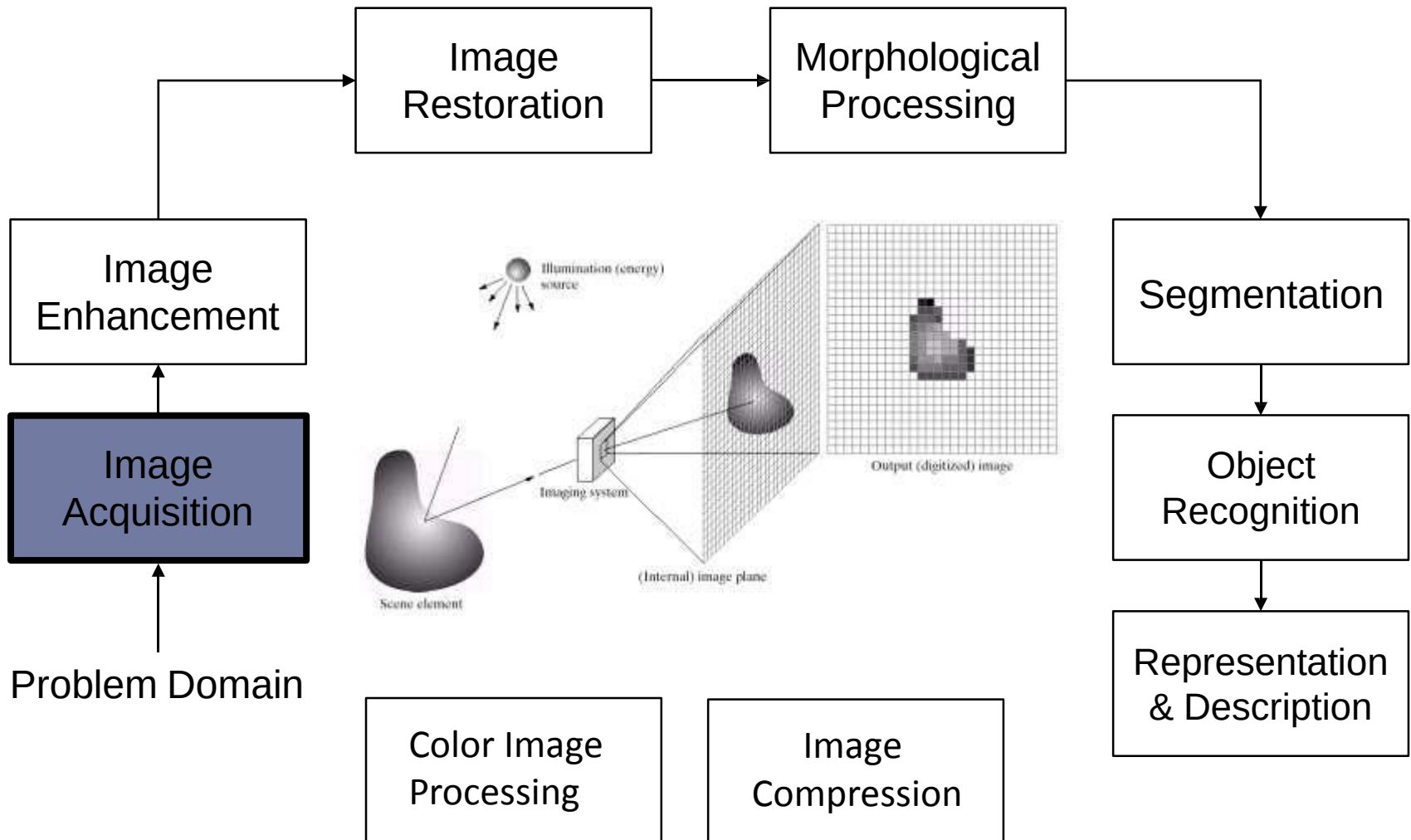


Image Enhancement

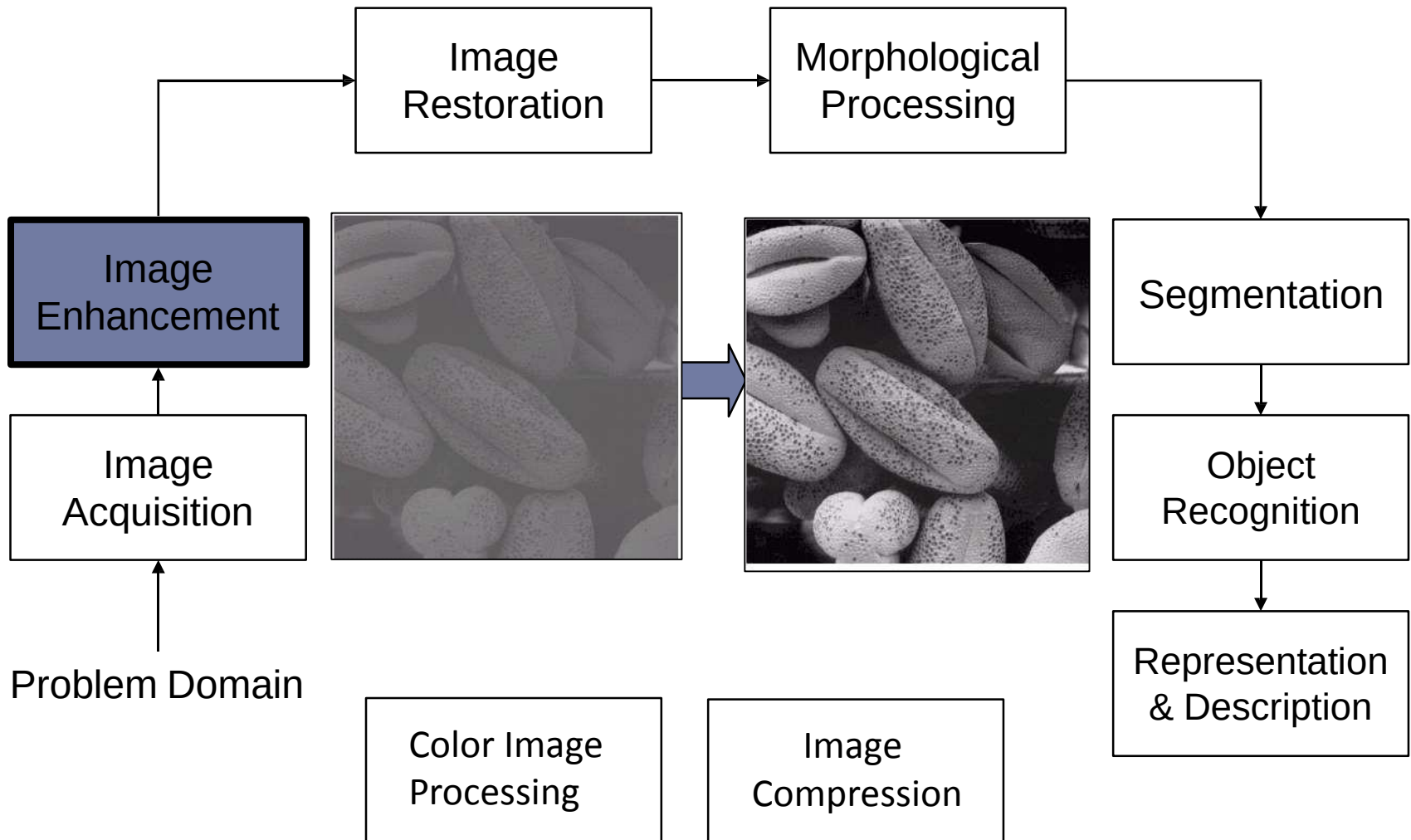
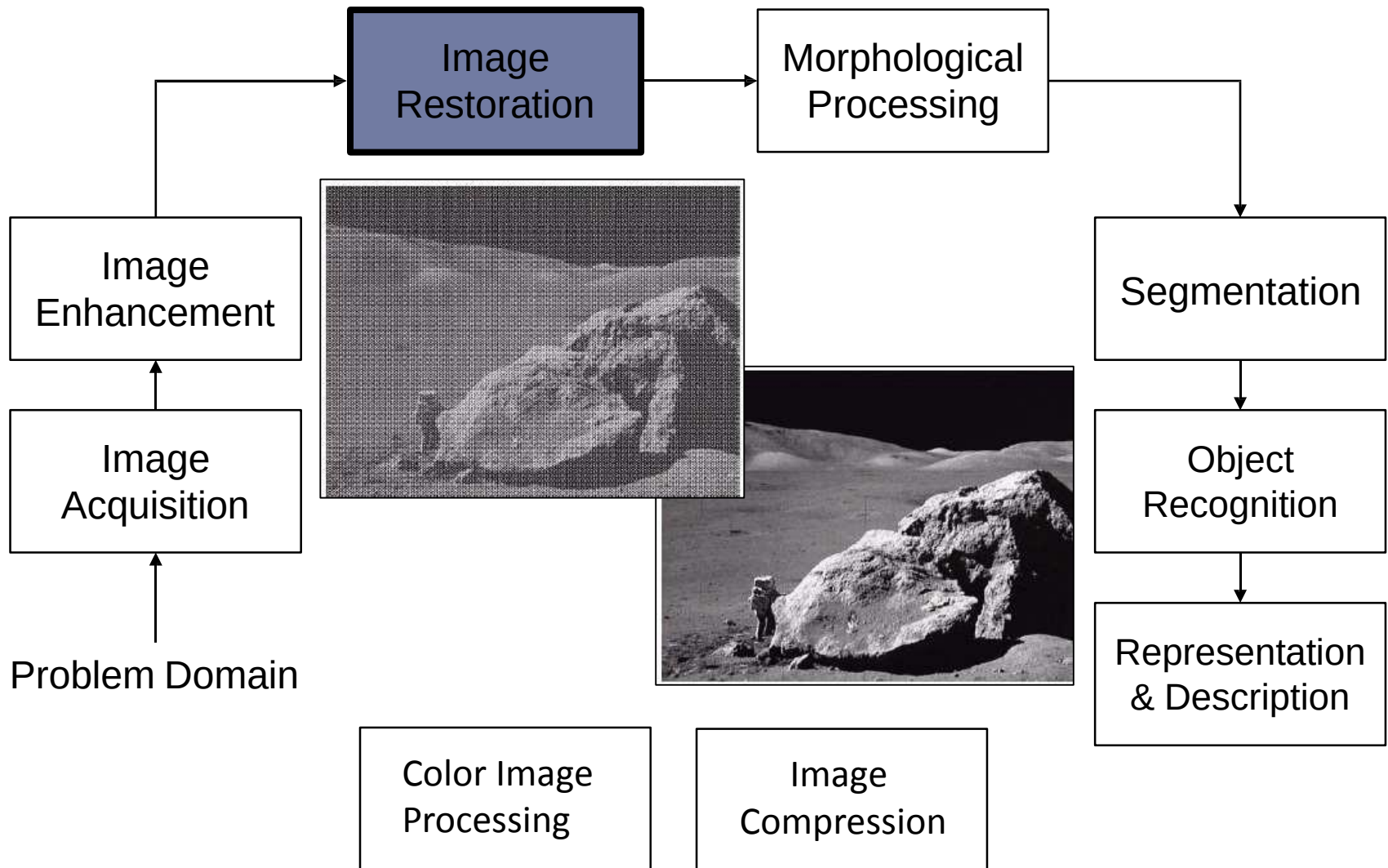
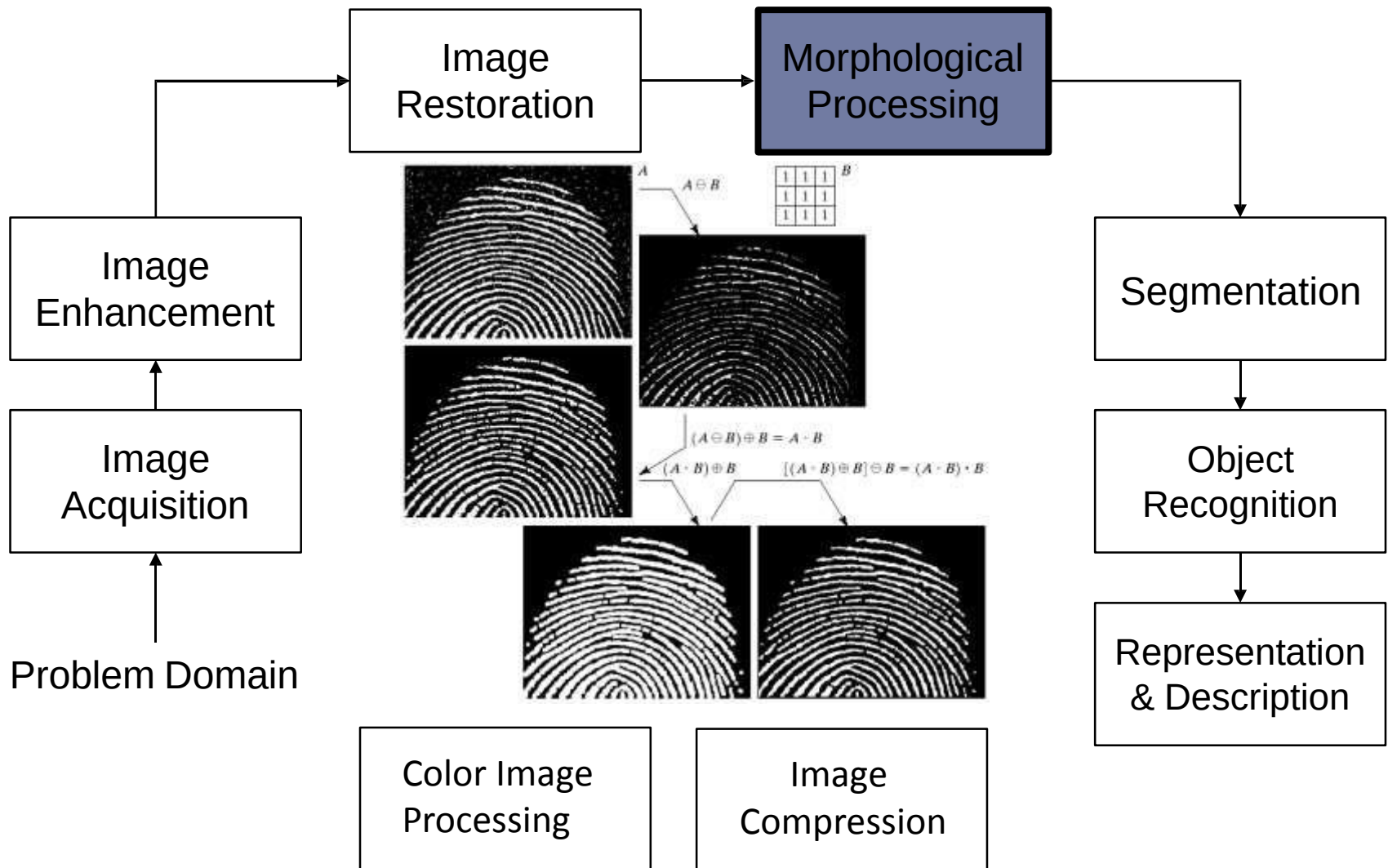


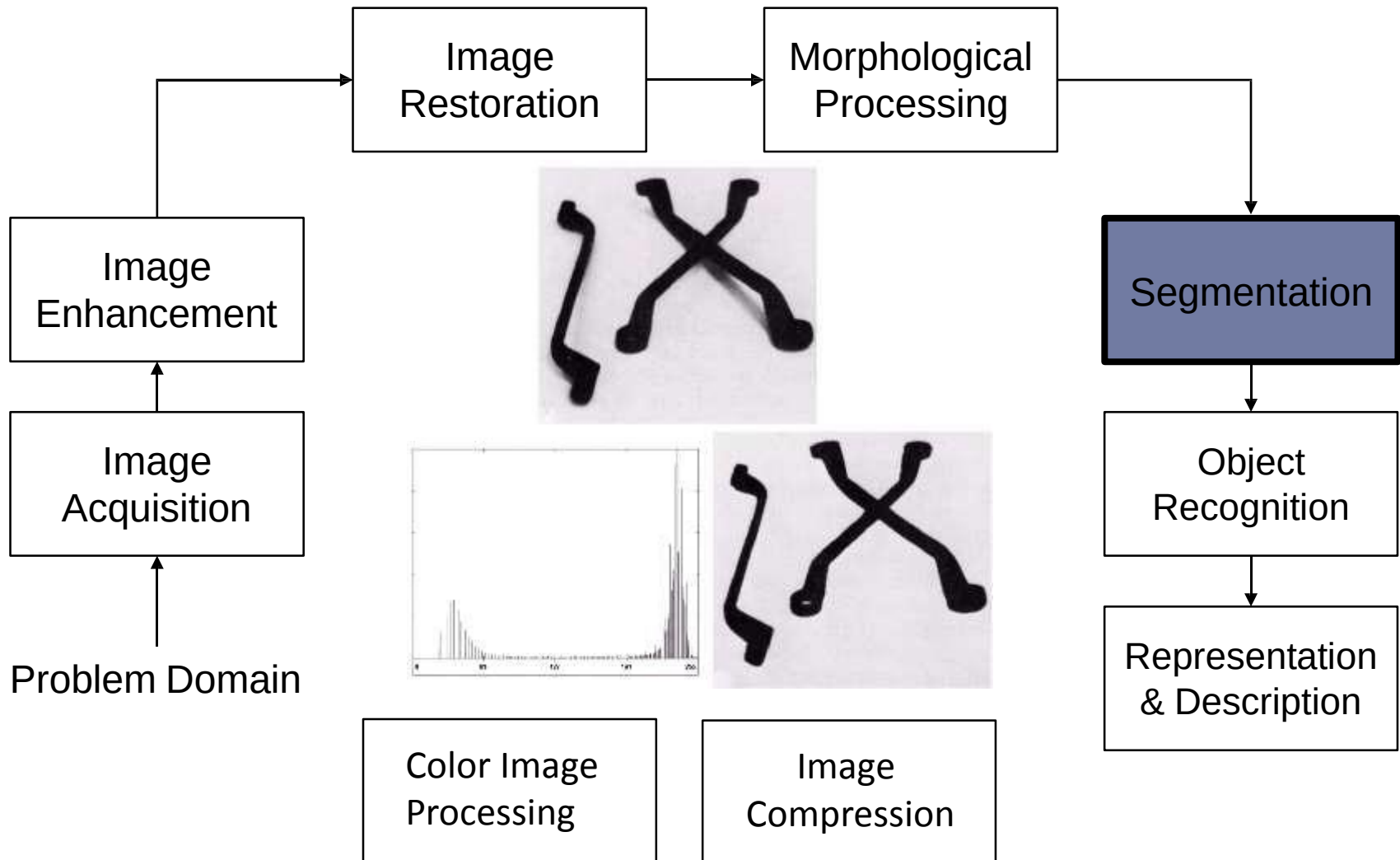
Image Restoration



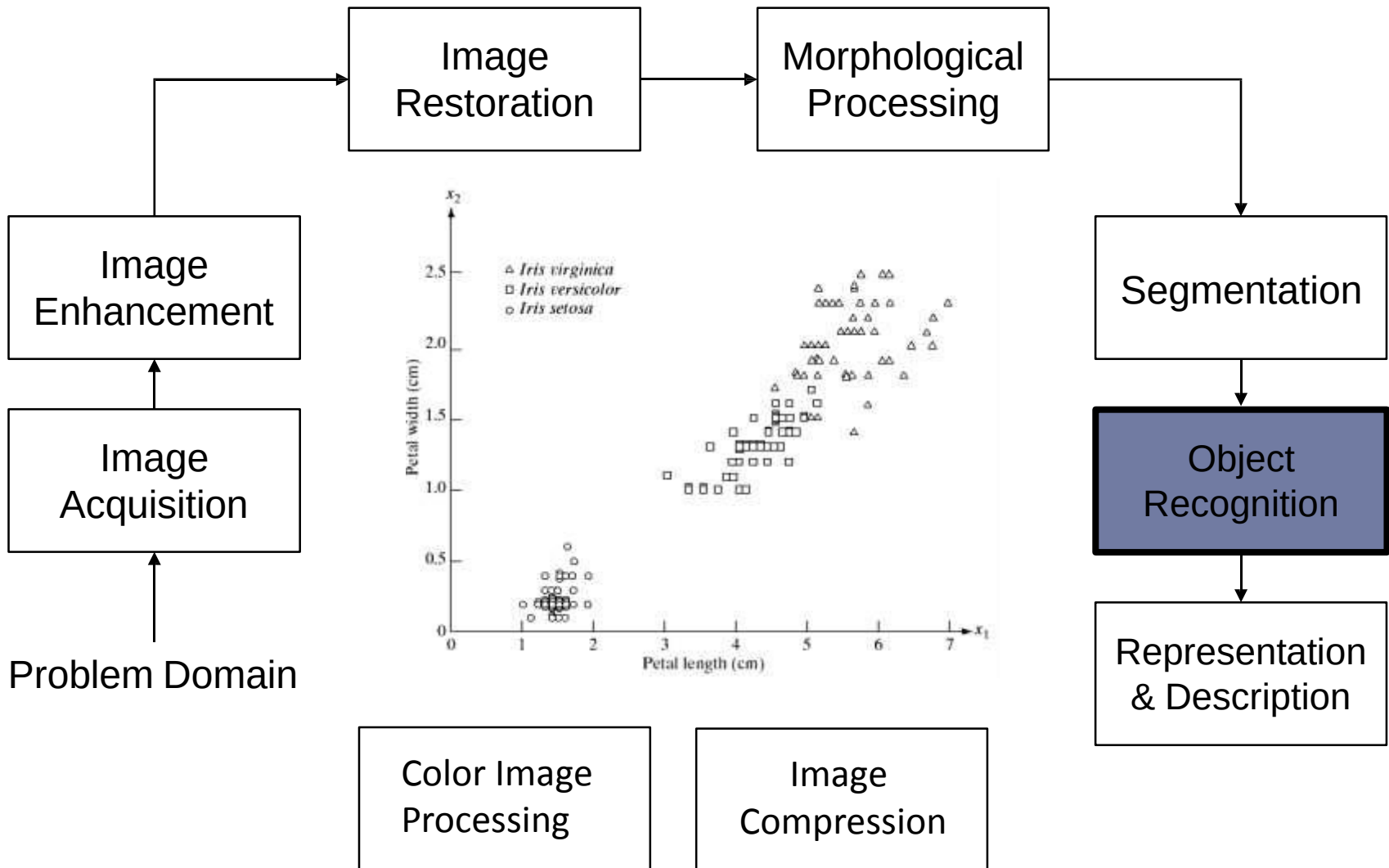
Morphological Processing



Segmentation



Object Recognition



Representation and Description

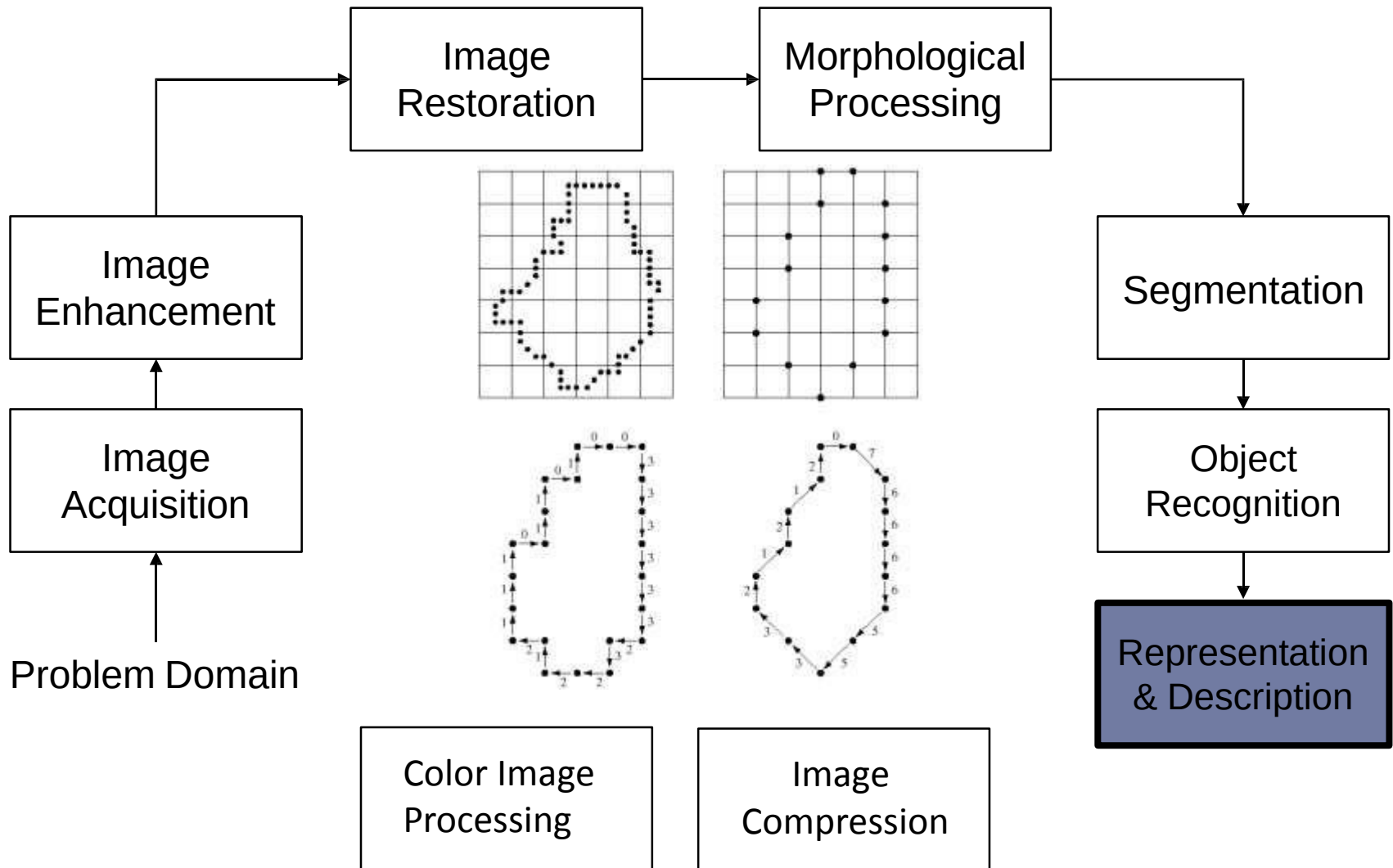
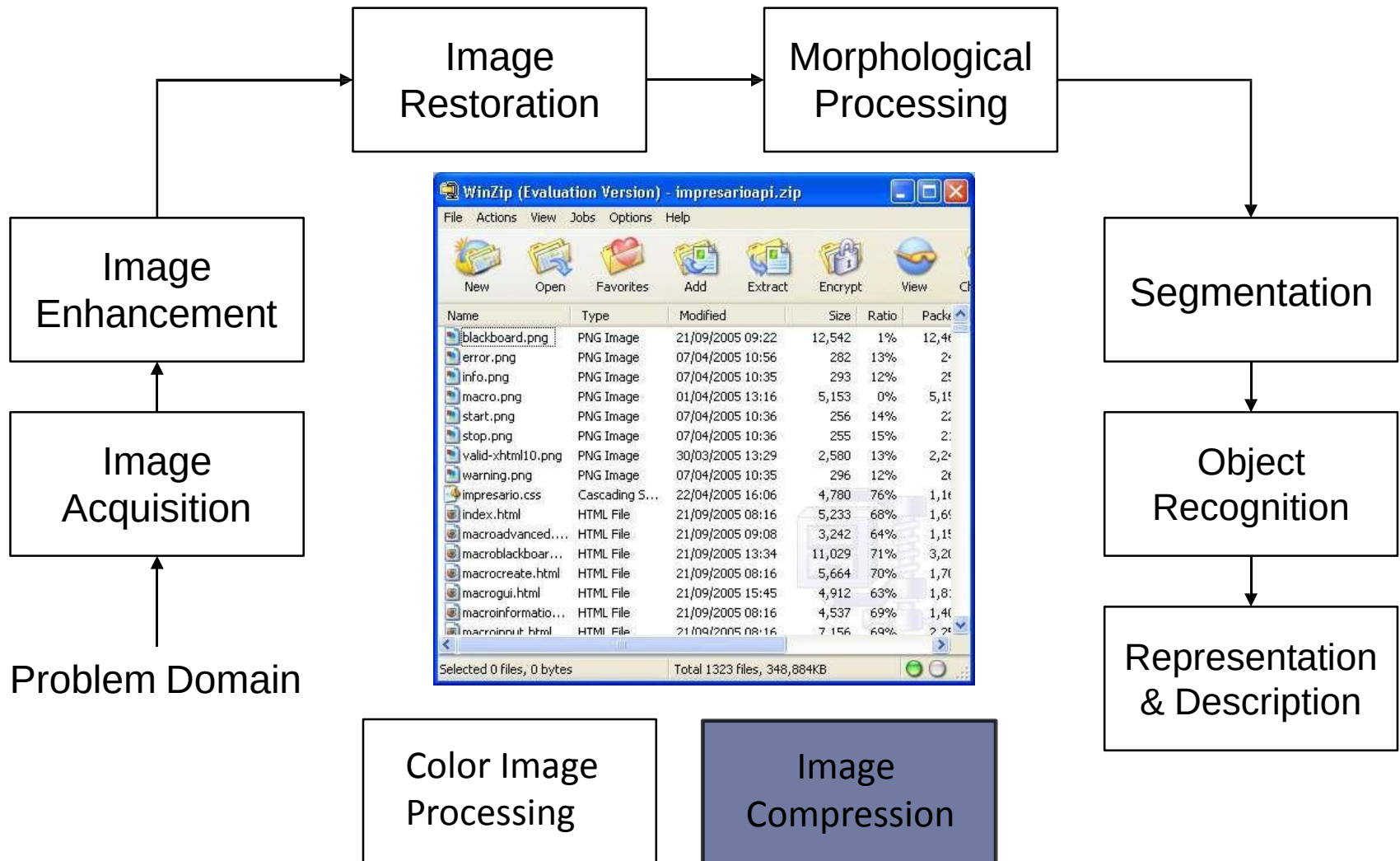
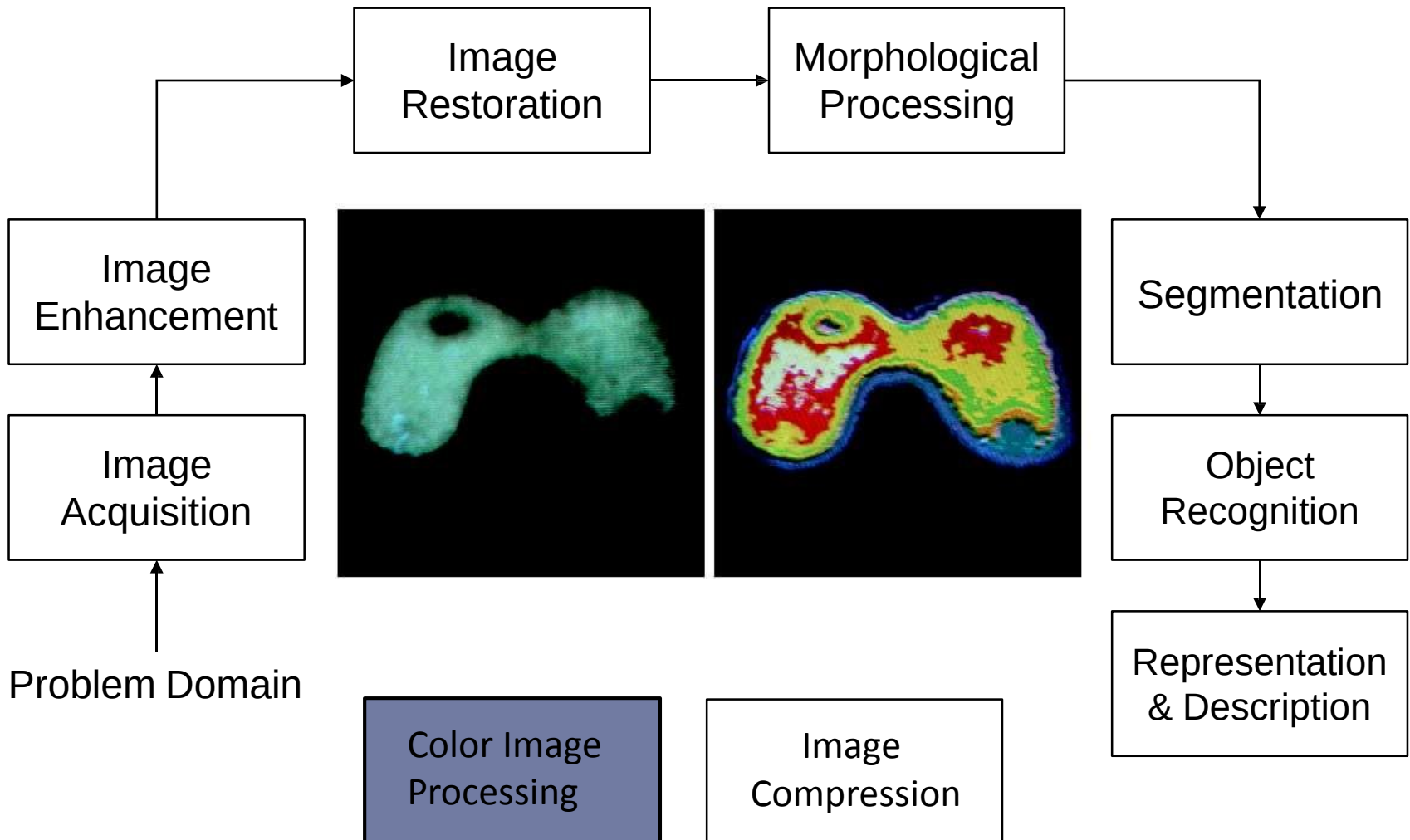


Image Compression



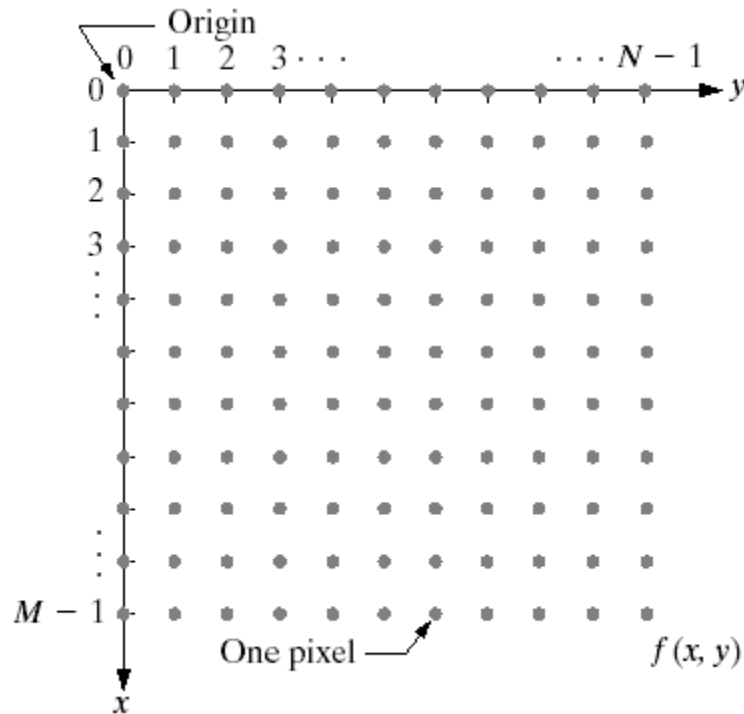
Color Image Processing



Components of an Image Processing System

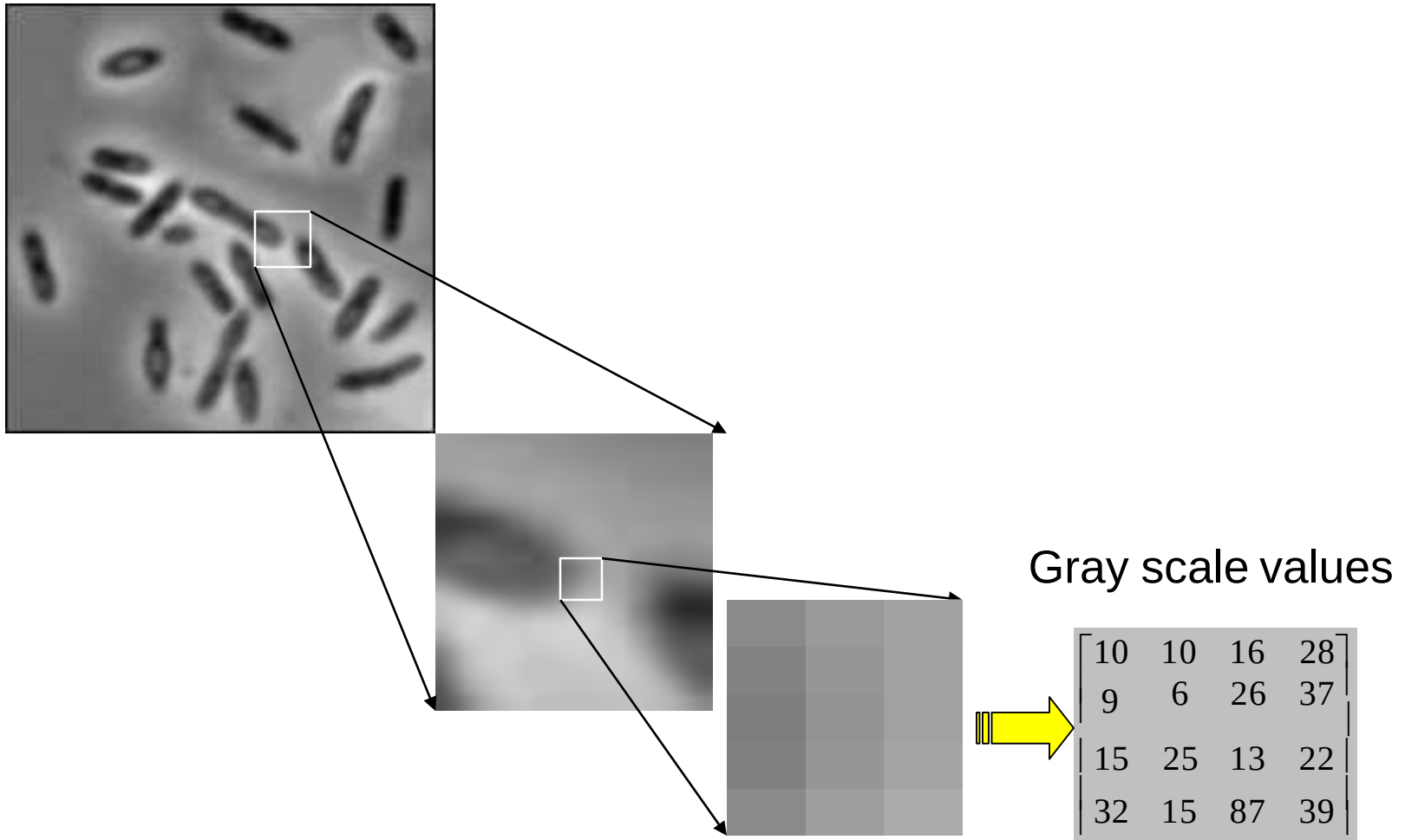
- Image sensor
- Specialized image processing hardware
 - This unit performs functions that require fast data throughputs (e.g., digitizing and averaging video images at 30 frames/s) that the typical main computer cannot handle.
- Computer
- Image processing software
- Mass storage
- Image display
- Hardcopy
- Networking

Representing Digital Images

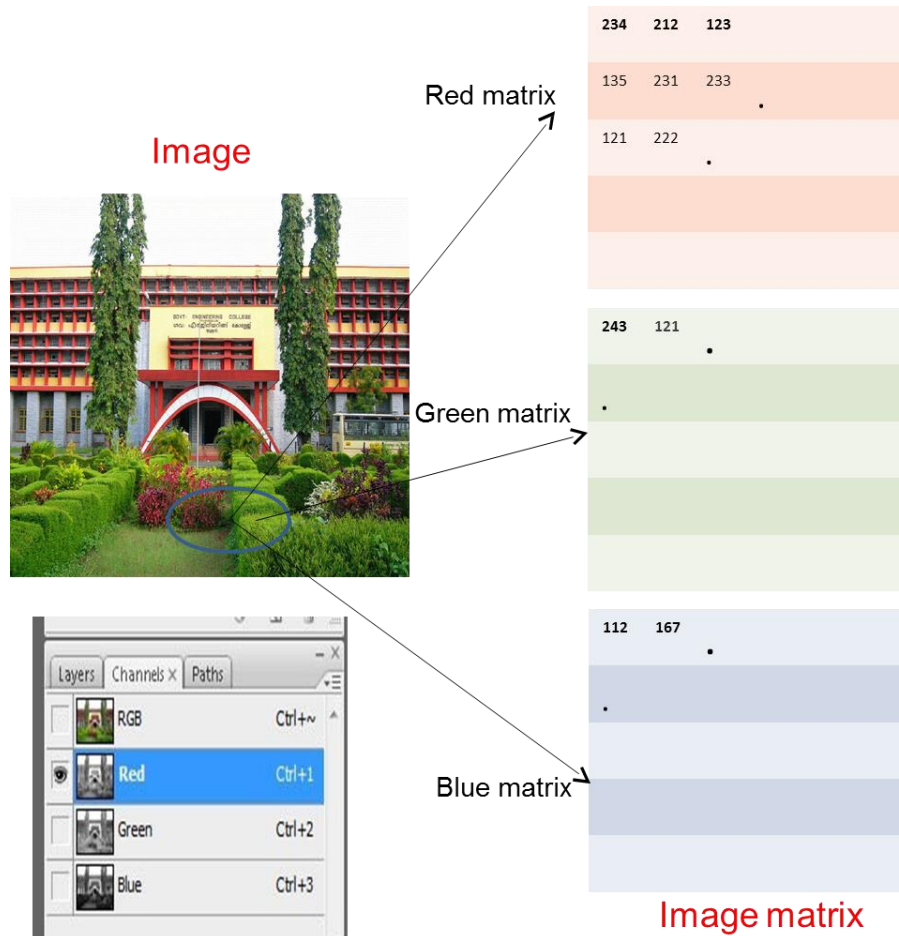


An image with size $M \times N$ and L ($=2^k$) discrete gray level for each pixel, the number of bits required to store a digitized image is $M \times N \times k$.

Monochromatic Image



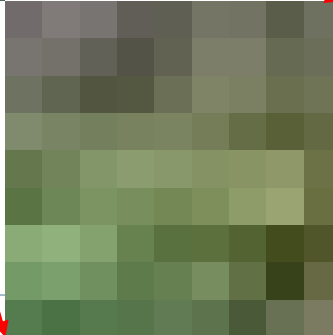
Color Image



Color Image



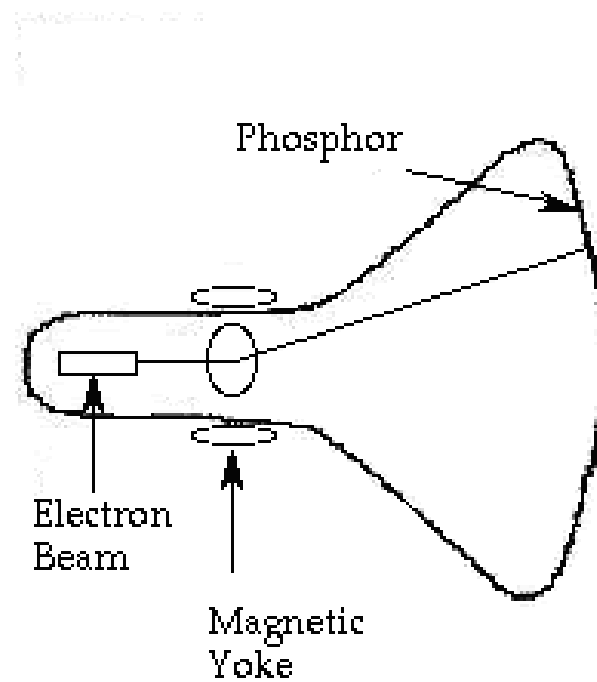
Each pixel contains a vector representing red, green and blue components



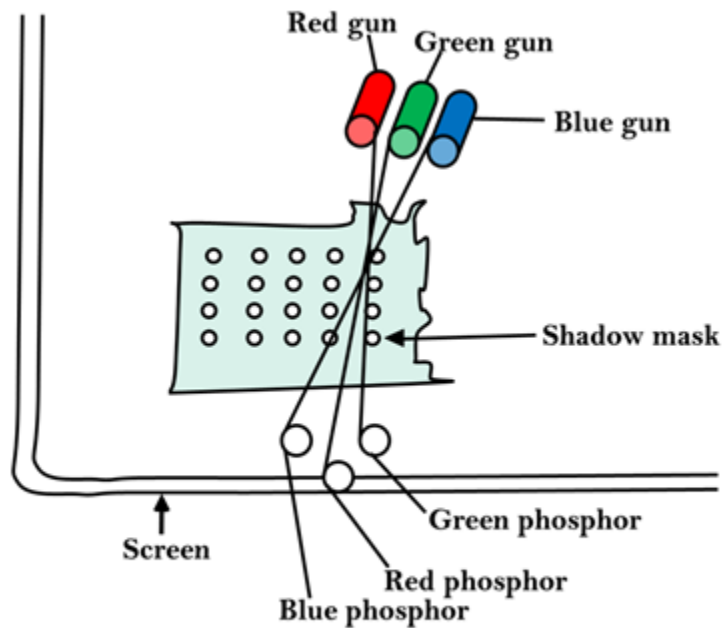
RGB components

10	10	16	28		
9	65	70	56	43	
15	32	99	70	56	78
32	21	60	90	96	67
	54	85	85	43	92
		32	65	87	99

Display Device (Gray)



Display Device (Color)



The Shadow mask CRT